HW_Assignment_2_Fall2026

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ENEL 573 Computer Networks: HW Assignment 2 (Fall 2026)
(Due on Oct 13th, at 11:59pm MST)

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Problem 1 (20 points): In modern packet-switched networks, the source host segments long, application-layer messages (for example, an image or a music tile) into smaller packets and sends the packets into the network. The receiver then reassembles the packets back into the original message. We refer to this process as message segmentation. Figure 1 illustrates the end-to-end transport of a message with and without message segmentation. Consider a message that is $7.5*10^6$ bits long that is to be sent from source to destination in Figure 1. Suppose each link in the figure is 1.5 Mbps. Ignore propagation, queuing, and processing delays.

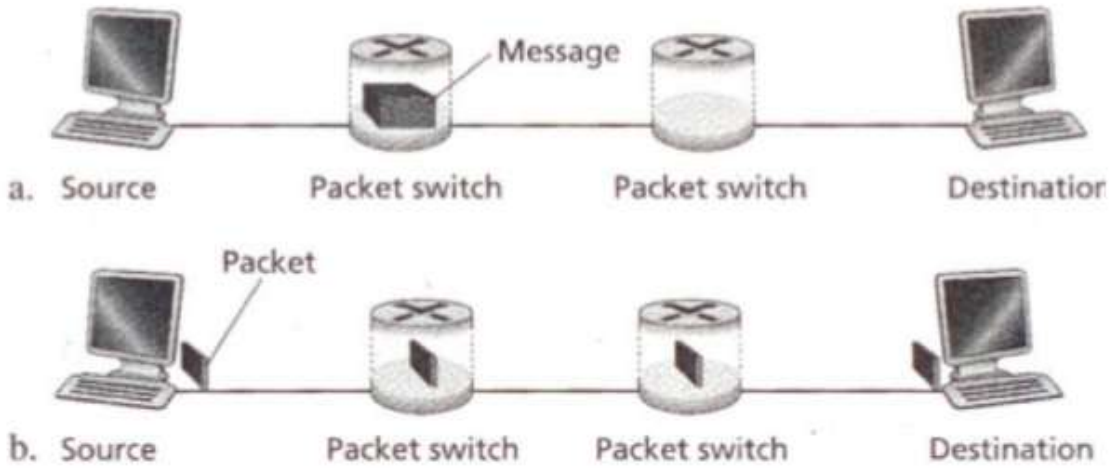


Figure 1. End-to-end message transport: (a) without message segmentation; (b) with message segmentation

(a) Consider sending the message from source to destination without message segmentation. How long does it take to move the message from the source host to the first packet switch? Keeping in mind that each switch uses store-and-forward packet switching, what is the total time to move the message from source host to destination host? **(5 points)**
(b) Now suppose that the message is segmented into 5,000 packets, with each packet being 1,500 bits long. How long does it take to move the first packet from source host to the first switch? When the first packet is being sent from the first switch to the second switch, the second packet is being

sent from the source host to the first switch. At what time will the second packet be fully received at the first switch? **(5 points)**

(c) How long does it take to move the file from source host to destination host when message segmentation is used? Compare this result with your answer in part (a) and comment. **(5 points)**

(d) Discuss the drawbacks of message segmentation. **(5 points)**

a) $d_{trans} = \frac{L}{R} = \frac{7.5 \times 10^6}{1.5 \times 10^6} = 5s$

$t_{total} = 3 \times 5s = 15s$

b) $d_{trans} = \frac{1500}{1.5 \times 10^6} = 1ms$ second packet is fully received at switch 1 at 2ms

c) $t_{total} = (N + links - 1)t = (5000 + 3 - 1)(1 \times 10^{-3}) = 5.002\,s$

segmentation is 3x faster because of pipelining. All links transmit in parallel.

d) 1. each packet needs extra header overhead
2. reassemble packets

Problem 2 (20 points): A communication line capable of transmitting at a rate of 50 Kbits/sec will be used to accommodate 10 sessions each generating Poisson traffic at a rate 150 packets/min. Packet length are exponentially distributed with mean 1000 bits. For each session, find the average number of packets in queue, the average number in the system, and the average delay per packet when the line is allocated to the sessions by using:

i) 10 equal-capacity time-division multiplexed channels; **(10 points)**

ii) statistical multiplexing. **(10 points)**

$\lambda = 150\,packets/min = 2.5\,packets/s$

$\frac{1}{\mu} = 1000\,b$

i) $\frac{50\,kbps}{10} = 5\,kbps$ for each channel

$\mu = \frac{5000\,bps}{1000\,b} = 5\,packets/s$

$\rho = \frac{\lambda}{\mu} = \frac{2.5\,packets/s}{5\,packets/s} = 0.5$

$N_Q = \frac{\rho^2}{1-\rho} = \frac{0.5^2}{1-0.5} = 0.5\,packet$

$N = \frac{\rho}{1-\rho} = \frac{0.5}{1-0.5} = 1\,packet$

$W = \frac{N}{\lambda} = \frac{1}{2.5} = 0.4s$

ii) $\lambda = (10)(2.5) = 25\,packets/s$

$\mu = \frac{50000\,bps}{1000\,b} = 50\,packets/s$

$\rho = \frac{\lambda}{\mu} = \frac{25}{50} = 0.5$

$\frac{N_Q}{10} = \left(\frac{1}{10}\right)\frac{\rho^2}{1-\rho} = \frac{0.5^2}{10(1-0.5)} = 0.05\,packet$ ← packet per session

$\frac{N}{10} = \left(\frac{1}{10}\right)\frac{\rho}{1-\rho} = \frac{0.5}{10(1-0.5)} = 0.1\,packet$ ← packet per session

$W = \frac{N}{\lambda} = \frac{1}{25} = 0.04s$

Problem 3 (30 points): In an M/M/1 queue, customers arrive at the rate of $\lambda = 15$ per hour. What is the minimum server rate to ensure that

1. the sever is idle at least 10% of the time **(10 points)**
2. the expected value of the queue length is not to exceed 10? **(10 points)**
3. the probability that at least 20 people in the queue is at most 50%. **(10 points)**

$\lambda = 15/hr$

$\rho = \frac{\lambda}{\mu}$

$P_n = (1-\rho)\rho^n$

1.) $1 - \rho \geq 0.1$

$\rho \leq 0.9$

$\frac{15}{\mu} \leq 0.9$

$16.67/hr \leq \mu$

2.) $N_Q \leq 10$

$\frac{\rho^2}{1-\rho} \leq 10$

$\frac{\left(\frac{15}{\mu}\right)^2}{1-\frac{15}{\mu}} \leq 10$

$\mu^2 - 15\mu - 225 \geq 0$

$\mu \geq 16.37/hr$

3.) $P(n \geq 21) \leq 0.5$

$\sum_{n=21}^{\infty} (1-\rho)\rho^n \leq 0.5$

$\rho^{21} \leq 0.5$

$\frac{15}{\mu} \leq \sqrt[21]{0.5}$

$15.5/hr \leq \mu$

$\sum_{n=N}^{\infty} \rho^n = \frac{\rho^N}{1-\rho}$

Problem 4 (30 points):

a) Compute the even parity bits for the data bits: 1 1 0 1 0 1 1 1 and 1 0 0 0 1 0 1 0 **(5 points)**

b) Compute the odd parity bits for the data bits: 1 1 0 1 0 1 1 1 and 1 0 0 0 1 0 1 0 **(5 points)**

c) Transform the following bit string (D) into a 3 x 8 matrix and compute the two-dimensional even parity EDC bits. D = 1 1 0 1 0 0 1 0 1 1 1 0 0 0 1 1 0 0 1 1 1 0 1 1 **(10 points)**

d) Assume that a sender transmits <D, EDC> as shown and computed in c). Next, assume that the receiver receives D' and EDC' as follows:

D' = 1 1 0 1 0 0 1 0 1 1 1 1 0 0 1 1 0 0 1 1 1 0 1 1; EDC' = EDC, Does the received bit stream have errors? If yes, how does the receiver detect or correct the errors and how many errors can be detected or corrected? **(10 points)**

a) 11010111 parity bit 1

10001010 parity bit 0

b) 11010111 parity bit 0

10001010 parity bit 1

c)

11010010	0
11100011	1
00111011	1
00001010	0

d)

11010010	0
11100011	1 ← error
00111011	1
00001010	0

↑ error

error @ row 2 column 4

to fix: flip the bit

1 error can be corrected

2-3 errors can be detected

4+ errors can go undetected (not reliable)

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